Evaluating the Ethical, Clinical, and Regulatory Implications of Large Language Models in Healthcare

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Significance & Problem

Large language models (LLMs) promise to revolutionise clinical practice, administration, and patient engagement while accelerating research and drug discovery.

However, their adoption raises concerns about data quality, hallucinations, bias and equity, privacy and transparency.

Our study examines this tension, seeking to ensure that efficiency gains do not compromise safety or ethics.

Research Question & Objectives

How can healthcare systems harness LLMs' potential while mitigating clinical, ethical and regulatory risks?

Aims & Objectives

- Map current LLM applications in healthcare
- Analyse limitations, biases and risks
- Examine regulatory frameworks (EU AI Act, MDR/IVDR)
- Develop a conceptual governance framework
- Consider adoption pathways for regulators and providers

APPLICATIONS & BENEFITS

Clinical care

Decision support, diagnostics (e.g., Med-PaLM 2 achieving near-expert performance)

Administrative tasks

Summarisation & documentation to reduce clinician workload

Patient engagement

Conversational tools for triage, education and personalised guidance

Research & drug development

Acceleration of literature reviews and molecule discovery

Challenges & Biases

Hallucinations & factual errors

Cognitive biases (suggestibility, anchoring, confirmation, framing)

Opaque reasoning & interpretability issues

Equity harms & marginalised populations



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Policy & Regulatory Frameworks

EU AI Act (2024/1689) classifies AI by risk: prohibited, high-risk, limited risk

High-risk AI (e.g., medical devices) requires conformity assessments, risk management, transparency, human oversight and logging

Healthcare organisations must ensure staff AI literacy, data control and ongoing monitoring

Other regulations: MDR/IVDR and GDPR complement AI Act – privacy impact assessments and conformity assessments remain mandatory

Methodology

DESIGN

Narrative synthesis of peer-reviewed articles (2020–2025), policy documents and guidelines on LLMs in healthcare

INCLUSION/EXCLUSION CRITERIA

Include English-language sources covering applications, risks, policies or frameworks for LLMs in healthcare; exclude preprints without peer review and news articles

CODING & SYNTHESIS

Two researchers independently apply thematic coding; disagreements resolved via discussion; maintain audit trail to ensure rigour and transparency

RESEARCHER BIAS MITIGATION

Reflective journaling and triangulation across sources reduce interpretation bias

Data Analysis & Evaluation

Thematic Categories

Applications & benefits

Risks & biases

Regulatory & ethical frameworks

Governance & adoption pathways

Evaluation Strategy

Synthesise findings using narrative summary; compare evidence strength and diversity; identify gaps and contradictions; align observations with ACUTE guidelines and health equity frameworks

Quality Appraisal

Assess methodological quality using adapted CASP checklists; highlight biases and limitations of included studies

Ethical Considerations

Secondary research – no direct patient data, but ethical review ensures responsible conduct

Critical appraisal of sources to minimise misinterpretation and amplifying biases

Reflexivity and transparency in coding decisions to mitigate researcher bias

Consider potential misuse or overreliance on LLM outputs; emphasise human oversight

Proposed Governance Framework

Implementation Guidelines

Retrain or adapt models using local datasets (retrieval-augmented generation)

Deploy models within secure, hospital-hosted environments to protect privacy

Continuous monitoring and auditing aligned with regulatory requirements

Core Principles

Accuracy & consistency of outputs (A,C)

Unaltered semantics & traceability (U,T)

Ethical compliance & oversight (E)



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Adoption & Implementation

Stakeholder Engagement

Collaborate with clinicians, ethicists, patients and regulators in co-designing frameworks

Policy Alignment

Map framework elements to AI Act requirements; integrate MDR/IVDR and GDPR provisions

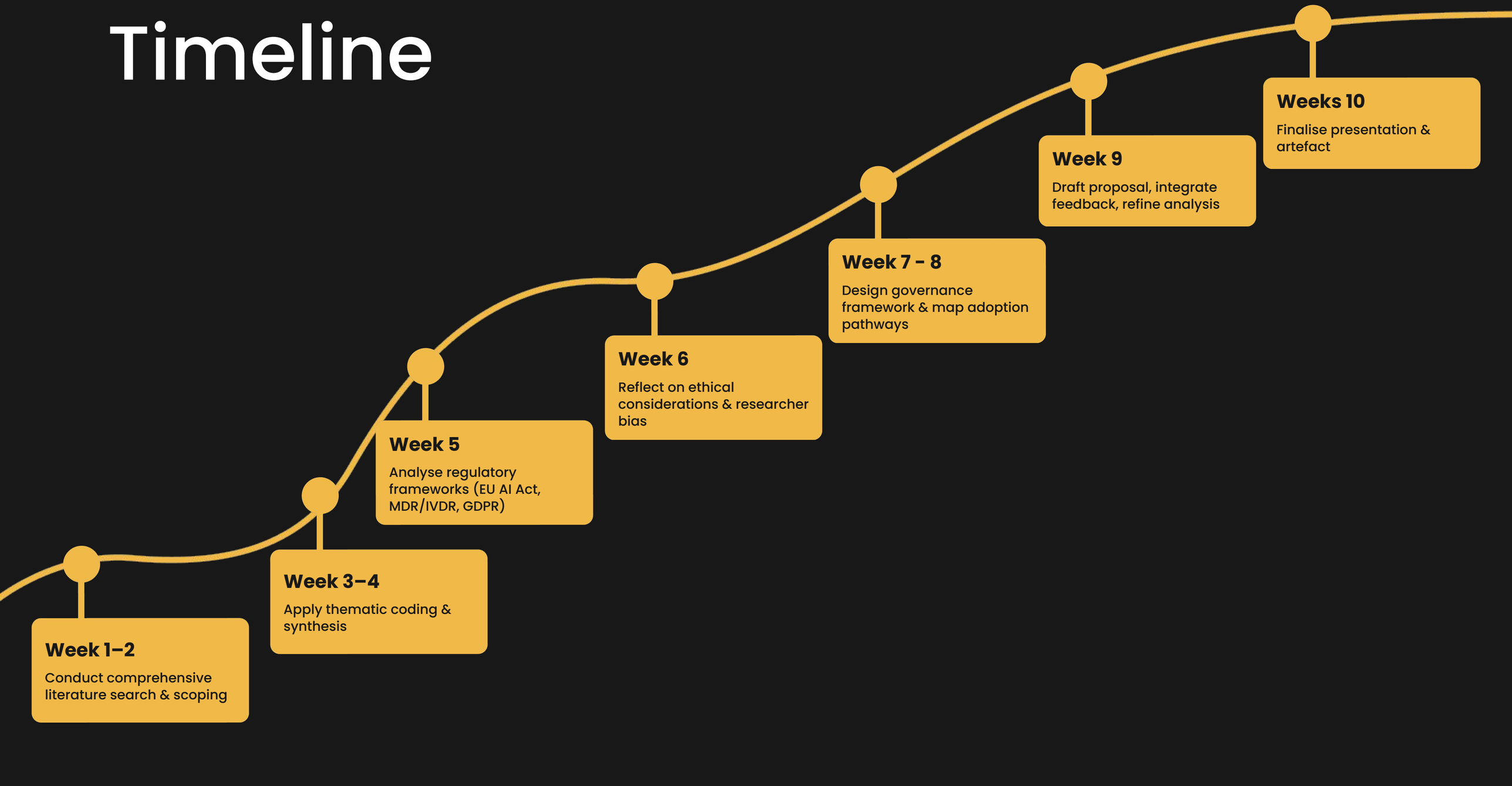
Capacity Building

Develop AI literacy programmes for healthcare staff and inform patients about AI use

Monitoring & Continuous Improvement

Establish feedback loops and post-deployment auditing; encourage open science and reporting

Timeline


Week 1–2
Conduct comprehensive literature search & scoping
Week 3–4
Apply thematic coding & synthesis
Week 5
Analyse regulatory frameworks (EU AI Act, MDR/IVDR, GDPR)
Week 6
Reflect on ethical considerations & researcher bias
Week 7 - 8
Design governance framework & map adoption pathways
Week 9
Draft proposal, integrate feedback, refine analysis
Weeks 10
Finalise presentation & artefact

Expected Contribution

Academic Contribution
Provide a holistic synthesis of clinical, ethical and regulatory dimensions of LLMs, highlighting gaps and opportunities for future research

Practical Contribution
Deliver a conceptual governance framework aligned with EU AI Act and healthcare realities, offering implementable guidelines for regulators and providers

Societal Impact
Promote safe, equitable adoption of LLMs to improve patient outcomes and maintain public trust

Conclusion

LLMs offer unprecedented opportunities to enhance healthcare delivery, but uncritical adoption risks harm.

Our proposal integrates applications, risks, policy and ethics into a governance framework to guide safe deployment.

Stakeholder collaboration and continuous evaluation are essential to harness benefits while protecting patients and upholding trust.

References

Amann, J., et al. (2020). Explainability for artificial intelligence in healthcare. Journal of Artificial Intelligence in Medicine.

BMC Editorial Board (2024). Navigating the integration of large language models in healthcare: challenges, opportunities and implications under the EU AI Act.

Canadian Agency for Drugs and Technologies in Health (2025). 2025 Watch List: Artificial Intelligence in Health Care.

Gerhilde van Leeuwen, K., et al. (2025). The AI Act: responsibilities and obligations for healthcare professionals and organisations. Diagnostic and Interventional Radiology.

Erlan, Y., et al. (2025). Large language models in medicine: applications, challenges and future directions. *International Journal of Medical Sciences*

Mahajan, A., et al. (2025) 'Cognitive bias in clinical large language models', npj Digital Medicine, 8, p. 428.

Singhal, K., et al. (2023). Towards expert-level medical question answering with large language models. arXiv preprint.

Pfohl, M., et al. 2024) 'A toolbox for surfacing health equity harms and biases in large language models', Nature Medicine, 30, pp. 3590–3600.

Workum, J.D., van de Sande, D., Gommers, D. and van Genderen, M.E. (2025) 'Bridging the gap: a practical step-by-step approach to warrant safe implementation of large language models in healthcare', Frontiers in Artificial Intelligence, 8, 1504805.